

LAB RESOURCE UTILIZATION PLATFORM

Document 2 Problem Statement

Infosys Springboard Internship

1. Existing System

Many educational institutions manage laboratory resources manually using paper registers or spreadsheets. Equipment booking, maintenance tracking, and utilization monitoring are handled separately, making the process time-consuming and prone to errors.

2. Problems in the Existing System

- Manual equipment booking leads to scheduling conflicts.
- No centralized database for managing laboratory resources.
- Difficulty in tracking equipment utilization.
- Maintenance records are difficult to maintain.
- Lack of real-time equipment availability information.
- Limited resource sharing between departments or institutions.
- Manual report generation consumes significant time.

3. Proposed Solution

The Lab Resource Utilization Platform provides a centralized web-based solution for managing laboratory resources. It enables secure user authentication, equipment management, online booking, maintenance scheduling, utilization tracking, notifications, and report generation through role-based access control.

4. Objectives of the Proposed System

- Automate laboratory resource management.
- Provide real-time equipment availability.
- Reduce booking conflicts.
- Maintain maintenance history.
- Improve equipment utilization.
- Support secure role-based access.
- Generate utilization and maintenance reports.

5. Benefits of the Proposed System

- Improves operational efficiency.
- Reduces manual work.
- Enhances transparency.
- Supports better decision-making.
- Ensures effective laboratory resource utilization.

6. Conclusion

The proposed system overcomes the limitations of manual laboratory management by providing an efficient, secure, and centralized platform for resource booking, maintenance, utilization tracking, and reporting.